

BIF 101 (Introduction to Bioinformatics):

Mcqs for Final Term

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Learn with Amina Zee

1. All organisms are composed of ____.

Cells

2. Cells are the ____ unit of life.

Basic

3. And comes from the ----- cells

Pre-existing

4. The tendency of an atom to attract electrons towards itself is called as ____.

Electronegativity

5. Living tissues are ____ water

70

6. _____ kinds of macromolecules are present.

Four

7. Macromolecules are _____ of smaller monomers.

Polymers

8. Cells are _____ to maintain area to volume ratio.

Small

9. Plasma membrane _____ Traffic.

Regulate

10. Nucleoid region contain _____.

DNA

11. An octagon protein complex surrounds _____.

Nuclear pore

12. Protein fibril made _____.

Cage like structure

13. Molecular biology grew from the biochemistry and _____.

Disciplines of genetics

14. In the present age it is in the midst of _____ scientific revolution.

Two

15. Synthesis of urea by ____ in ____.

Friedrich Wohler, 1828

16. The First enzyme was ____.

Diastase

17. Diastase was discovered by ____ in ____.

Anselme payen, 1833

18. Mendel experiment on garden pea by ____ in ____.

Gregormendel, 1865

19. Chromosome theory of inheritance by ____ in ____.

Thomas and Hunt morgan, 1910

20. Discovery of DNA by ____ in ____.

Friedrich meisner, 1869

21. Cell free fermentation by ____ in ____.

Eduard Buchner, 1897

22. Crystallized enzyme urease by ____ in ____.

James B summer, 1926

23. Physical evidence of Recombination by ____ in ____.

Barbara mac clintock and harriet crieghton, 1931

24. One gene one enzyme was by _____ in ____.

Gergor beadle and Edward tatum, 1941

25. James Watson and francis crick discover _____ in ____.

Double helix structure of DNA, 1953

26. DNA is hereditary materials demonstrate by ____ in ____.

Oswald avery and his colleagues, 1944

27. DNA was confirmed by _____ In ____.

Alfred harshey and Martha chase, 1952

28. Francis laid out the Central dogma of molecular biology in ____.

1957

29. Mathew meslson and franklin stahl proved that replication of DNA is ____

Semi -Conservative

30. Marshall nerinberg and Gobind Khorana cracked the code in ____.

1960

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31. Codon stands for ____

One amino acid

32. three bases constitute a code word called ____

Codon

33. In 1970 ____ was discovered.

Gene cloning

34. Human genome project was launched in_____

1990

35. Human Genome Project completed in_____

2003

36. Nuclei acid are important groups of-_____

Bi molecules

37. Nuclei acid is responsible for transmission and _____

Storage

38. Nucleic acid are _____ compounds.

Polymeric

39. Repeating unit of nucleic acid is _____

Nucleotides

40. Nucleic acid consists of _____

RNA, DNA

41. DNA is a polymer of _____

Deoxyribonucleotides

42. Deoxyribonucleotides composed of _____ components.

Three

43. DNA composed of:

Phosphoric acid, nitrogenous base, deoxyribose sugar

44. A molecule containing all these three molecules is called _____

Nucleotide

45. While a molecule without the phosphate group is called ____

Nucleoside

46. A Nucleotide composed of :

Nitrogenous base, sugar and Phosphoric acid

47. A Nucleoside contains :

Nitrogenous base and a sugar

48. There are ____ Types of deoxyribonucleotides.

Four

49. Four types of Deoxyribonucleotides are ____.

Deoxy-thymidine, Deoxy-cytidine, Deoxy-guanosine, Deoxy-adenosine

50. These four structures make up the structural unit of ____

DNA

51. Successive nucleotides are joined together by ____

Phosphodiester bond

52. Base composition of DNA varies from one ____ to another.

Species

53. Study of genes, heredity and chromosomes is called ____

Genetics

54. Genetic makeup of organism is called ____

Genotype

55. Physical appearance of organism is called ____

Phenotype

56. First proteins sequenced in ____ by ____ method.

1951, Sanger and tupy

57. Atlas of protein in ____ by ____.

1960, Margret dayhoff and his colleagues

58. Genome informatics is the field in which Computational and ____ techniques are applied.

Statistical

59. E-coli is ____

Bacteria

60. S- Cerevisiae is a:

Yeast

61. C-elegans is a ____

Worm

62. D.Melanogaster is a ____

Fly

63. Danio rerio is a ____

Zebra fish

64. Mus Musculus is ____

Mouse

65. Homo sapiens is ____

You and me (Human)

66. Arabidopsis is ____

Plants

67. Prokaryotes are the organism whose genetic material is ____

Not enclosed in a nuclear membrane

68. First prokaryotic genome is ____

Hemophilus influenza

69. Prokaryotes have.... Genomes.

Simple

70. Eukaryotes have.....genomes.

Larger

71. Prokaryotes have ____ Genomes.

Simple

72. Eukaryotes have ____ genomes.

Larger

73. Eukaryotes have ____ in their protein coding.

Introns

74. Densely packed region is ____

Heterochromatin

75. Lightly packed region ____

Euchromatin

76. Densely heterochromatin ____

Take more color

77. Lightly euchromatin....

Take less color

78. Most common in children whose parents are first cousins....

12.5%

79. Humans have over..... genes.

25000

80. These genes may produce.....different proteins.

250000

81. We need help of ____ to store and recall the data.

Computers

82. Computers can ____ process this information.

Quickly

83. If you carefully search this online data with a bit of biology it is.....

Gold mine

84. Biology easily has.....exciting year's problems.

500

85. Life, disease and..... are better understood.

Evolution

86. ____ can be better developed after understanding molecular basis of life.

Drugs

87. _____ are systems biology are other research areas in bioinformatics.

Protein structure, protein-protein interaction

88. Bioinformatics has enabled us to generate ____ data and its utilization.

Omic

89. Bioinformatics is relatively ____ scientific area.

Young

90. Identification of hitherto unknown types are due to ____

Roles of RNA

91. Bioinformatics is full of ____

Challenges and opportunities

92. Accurate solution of protein structure is one of the ____

Toughest Problem.

93. Some medicines have ____ on patients.

Side effects

94. century of bioinformatics is

21st century

95. It is an era of personalized medicines and ____ of several common diseases.

Eradication

96. Genetic information flow from ____

DNA to RNA

97. All living cells ____ their DNA when they divide.

Replicate

98. During cell division each daughter cell receive

Copy of genome.

99. ____ is the process by which two identical copies of DNA are made from an original molecule.

Replication

100. Replication occurs in the cells to cell division.

Prior

101. Information does not Flow

From protein to DNA or RNA.

102. Flow of information from RNA DNA.

To Backwards

103. When hypothesis was adopted chromosomal DNA function as a template.

In 1953

104. RNA molecules are move in ____

Cytoplasm

105. Concept of central dogma occurs in ____

1953

106. Central dogma referred by ____

Francis Crick

107. RNA chains are acts as a template for the synthesis of ____

DNA

109. Transformation of information in DNA in those molecules is termed as ;

Central dogma

110. Central dogma is _____

DNA > RNA > Proteins

111. Proteins are used in constructing on _____

Cells

112. DNA and RNA molecules are built up from ____Types of molecules.

Four

114. Nucleotides are ____

Adenine, Cytosine, uracil and guanine

115. DNA has

A, T, C, G

116. RNA has ...

A, U, G, C

117. DNA is ____ Stranded.

Double

118. RNA is ____Stranded.

Single

119. RNA helps to produce....

Proteins

120. mRNA nucleotides forms a codon of _____

Three nucleotides

121. tRNA has ____ Nucleotides.

75 to 80

122. Conformation of tRNA allows it to bind ____

Ribosomes

123 ____ Are built up of proteins and carbohydrate.

Cells

124 Molecules are produced after ____ Of DNA into proteins.

Transformation

125. Adenosine pairs with ____

Thymine

126. Cytosine pairs with ____

Guanosine

127. Which process decodes the bases in DNA.

Transcription

128. In RNA Uracil only pairs with ____

Adenosine

129. DNA gives rise to ____.

RNA

130. There are ____ Types of Ribonucleotides.

Four

131. Ribonucleotides are depending on the types of which bases.

Nitrogenous

132. There are ____ types of ribonucleic acid.

Three

133. The ribonucleic acids are ____

mRNA, tRNA, rRNA

134. mRNA is the types of RNA that carries information from ____

DNA to proteins

135. Process of forming mRNA is called ____.

Transcription

136. mRNA maybe ____

Monocistronic

137. Length of mRNA is depend on ____

Length of gene

138. Which serve as adapter molecules in the process of protein synthesis?

tRNA

139. t RNA are Linked to an amino acid at one end by ____

Covalently bond

140) which are components of ribosomes?

rRNA

141. rRNA is the ____Material.

Predominant

142. rRNA constitutes about ____Of ribosomes.

60%

144. mRNA is ___stranded when it is formed from DNA.

Single

145. Conformation is mainly achieved by

Base stacking Interactions.

146. Protein coding region of mRNA is composed of non-overlapping string of codon called___

ORF

147. Translation starts at 5' and ends at

3' end

148. First and last codon of an ORF are called___

Start and stop codon

149. In bacteria start codon is usually

5' - AUG-3'

150. Eukaryotic cells always use ... as start codon.

5'-AUG-3'

151. Start codon has how many functions.

Two

152. Eukaryotic mRNA almost always contain a ___

Single ORF

153. Prokaryotic cells have mRNA always contain ___

Two or more ORF

154. mRNAs containing multiple ORFs are known as ___

Polycistronic RNAs

155) Those encoding a single ORF are known as ____

Monocistronic ORF

157. The RBS typically located

3-9 Bp.

158. The phenomenon of linked translation between overlapping ORFs is known as ____

Coupling translation

159. Heart of protein synthesis is ____

Translation

160. tRNA molecules are between

75 and 95 nucleotides

161. Dihydrouridine derived from ...

Uridine

162. Hypoxanthine plays a role in the process of

Codon

163. tRNA molecules to which an amino acid is attached are said to be ____

Charged

164. tRNAs that lack an amino acid are said to be ____

Un-charged

165. Acyl linkage is ____ bond.

High energy

166. Hydrolysis results in a large change in_____

Free energy

168. All aminoacyl-tRNA synthetases attach an amino acid to a _____ tRNA in how many steps.

Two

169. Two enzymatic steps are

Adenylation , tRNA charging

170. how many amino acids is attached to the appropriate _____ tRNA.

20

171. Most amino acid are specified by _____

More than one Codon

172. Which Is the macromolecular machine that directs the synthesis of proteins.

Ribosomes

173. Ribosome is larger and more _____ Than the minimal machinery.

Complex

174. Polymerizing amino acid is composed of at-least

Three amino acids

176. Translation takes place at a rate of Amino acid per second.

Only two to 20

177. In prokaryotes transcription and translational machinery are located in _____

Same Compartment

178. Ribosomes catalyzes a _____.

Single Chemical reaction

179) Ribosomes is composed of

Two Subassemblies

180. Subassemblies of RNA are known as _____

Large and small units

181. Large subunit of RNA contain

Peptide transferase center

182. Peptidyl responsible for the _____

Formation of peptides bond

183. Small units of RNA contain ...

Decoding center

184. Small and large units of RNA are buried within the intact of

Ribosomes

185. A-site is the binding site for

Aminoacylated-tRNA

186. P-site is the binding site for

peptidyl tRNA

187. Stop codons are recognized by proteins called _____

Release factors

189) which process involves coding of proteins by RNA at ribosomes.

Translation

190. Sequence of amino acid are linked by

Peptide bond

191. Proteins are polymers of....

Amino- acids

192. Proteins are range in size from

Small to Large

193. Twenty different proteins are called

Standard proteins.

194. Each type of protein has a ____ amino acid sequence.

Unique

195) Linus Pauling and Robert Corey carefully analyzed the ____

Peptide bond

196. Six atoms of peptide group are ____

Planar

197) In principle, ϕ and ψ can have any value between....

+180, -180

198. Secondary structure of protein refers to ____

Local conformation

199. Backbone of polypeptide chain extend into ____

Zig- zag Structure

200. Three dimensional arrangements of atoms of proteins are ____

Tertiary structures

201. A multi-subunit protein is also referred to as a ____

Multimer

202) A multimer with just a few subunits is called ____

Oligomer

203. A single subunit is called ____

Protomer

204. Oligomers can have either ____ Symmetry.

Rotational, Helical

205. The process of inexact matching while keeping in view the conserved residues is called sequence.....

Alignment

206. Pairwise sequence alignment helps to compare ____.

Two Sequence

207. Gaps are not counted in:

Identity

208. Multiple sequence alignment involves comparison of:

Three or more sequences.

209. Pairwise alignment either local or ____

Global

210. MSA can help to align ____

Multiple Sequences

211. Progressive alignment can help to perform ____

MSA

212) CLUSTAL is an ____

Online tool

213. Pairwise alignment is the alignment of ____

Two sequences

214. CLUSTAL is an online tool to perform ____

MSA

215. BLAST developed in ____

1990

216. BLAST is an basic ____

Local alignment tool

217. BLAST consist of how many types:

Two

218. Two main types of BLAST are ____

Proteins, nucleotides

219. Smith waterman perform ____

Local Alignment

220. Needleman waterman calculates ____

Global alignment

221. FASTA developed in which year ____

1998

221. First Genbank developed in ____

1982

222. Genbank is now under

NCBI

223. EMBL developed in ____

1980

224. DDBJ established in ____

1984

225. First protein sequenced was collected in ____

1951

226. Protein sequence was collected by ____

Sanger and Tuppy method

227. Atlas of protein sequenced by

Margrett and DAYhoff in 1960

228. Swiss prot is collaboration between ____

EBI and SIB

229. GEO is a ____

Public repository

230. Informatics in health care is called ____

Health informatics

Best of Luck

☺Thank you ☺